

FERDIANSYACH

PROJECT PORTFOLIO — FULLSTACK DEVELOPER & DATA ANALYST

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Depok, Indonesia | ianferdiansyach30@gmail.com | +62 888 6007 599 | [linkedin.com/in/ferdiansyach](https://www.linkedin.com/in/ferdiansyach) | github.com/ferdiansyach | ferdiansyach-portfolio.vercel.app

5

PRODUCTION PROJECTS

92%

ML MODEL ACCURACY

27+

TECHNOLOGIES USED

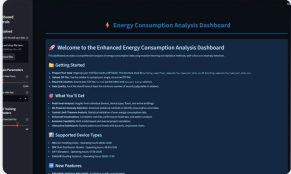
3

EXPERTISE DOMAINS

Full-Stack Developer & Data Analyst with a proven track record of building 4+ production-grade projects end-to-end using React, Next.js, Node.js, and Python. Experienced across interactive frontend architecture, backend REST APIs, and ML pipelines (LSTM, XGBoost) achieving 92% accuracy at Telkom Indonesia. This portfolio showcases projects with measurable impact and cross-domain technical expertise.

FEATURED PROJECTS

Sorted newest first



ML / DATA SCIENCE

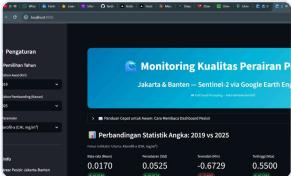
01 Smart Meter Analysis

2025

Code ↗

Python · Machine Learning · Data Analysis · Streamlit · XGBoost · LSTM · Pandas

- Engineered LSTM + XGBoost predictive models achieving 92% accuracy on 50,000+ smart meter data points for energy consumption forecasting.
- Built an end-to-end data pipeline (ingestion → preprocessing → deployment) with large-scale anomaly detection for Telkom Indonesia's techno-economic analysis division.
- Developed a real-time Streamlit dashboard for energy consumption visualization and model performance comparison, adopted directly by the internal team.



ML / DATA SCIENCE

02 Coastal Water Quality Monitoring

2025

Code ↗

Python · Google Earth Engine · Sentinel-2 · Landsat-8 · Streamlit · K-Means · Mann-Kendall · Remote Sensing

- Built a temporal coastal water quality monitoring system for Jakarta & Banten (2019–2025) using Google Earth Engine with multi-sensor Sentinel-2 & Landsat-8 data.
- Implemented K-Means clustering for water quality zoning and Mann-Kendall trend tests for statistical detection of parameter changes (turbidity, chlorophyll-a, TSS).
- Developed an interactive Streamlit dashboard for spatial & temporal water quality visualization, making satellite data interpretation accessible to non-technical stakeholders.



CMS / WORDPRESS

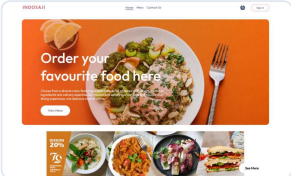
03 HIMASI UNAS Website

2024 – 2025

Code ↗

WordPress · Content Management · SEO · Plugin Management

- Developed the official student association website on WordPress, increasing traffic by 40% through structured, SEO-optimized content.
- Managed plugin ecosystem and content architecture, enabling non-technical members to self-manage content without compromising professional appearance.
- Created comprehensive website handover documentation, ensuring a smooth management transition to the next student association committee.



WEB DEV

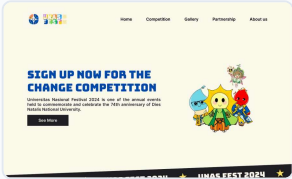
04 Indosaji E-commerce

2024

Code ↗

React · Node.js · Express.js · MongoDB · REST API · JWT Auth · Stripe

- Built a full-stack MERN e-commerce platform with Stripe payment gateway integration, processing real-time user transactions from client to server.
- Implemented React Context API for global state management (auth & shopping cart), ensuring consistent UX across all application views.
- Architected RESTful API with Node.js & Express backed by structured MongoDB schema, enabling secure admin/user role separation.



WEB DEV

05 Website UNAS FEST

2024 Code ↗

TypeScript · React · Tailwind CSS · Next.js

- Developed 10+ responsive components with TypeScript & Tailwind CSS, achieving a Lighthouse score of 90+ for performance and accessibility on the official festival portal.
- Engineered a comprehensive testing pipeline (Jest, RTL) reducing the production bug rate by 60% before a public release used by the entire university community.
- Collaborated in a multi-developer team using Git branching strategy and code reviews to maintain code quality under tight deadlines.

WORK EXPERIENCE

Business Intelligence & ML Intern — Telkom Indonesia — Infranexia Apr 2025 – Sep 2025

- Engineered a predictive AI model using LSTM and Gradient Boosting that achieved 92% accuracy in smart building energy consumption forecasting, directly supporting the techno-economic analysis division.
- Architected an end-to-end data pipeline from raw ingestion through preprocessing to model deployment (LSTM, XGBoost), processing 50,000+ data points and generating actionable insights for energy efficiency optimization.
- Developed a web-based internship management platform in collaboration with 3 cross-functional divisions, serving 100+ interns and reducing onboarding coordination overhead.

Computer Laboratory Assistant — Lab Data Monetize — Universitas Nasional Sep 2024 – Sep 2025

- Maintained and troubleshooted 30+ laboratory computer units, sustaining 98% device availability during operational hours and minimizing session downtime for 200+ students per semester.
- Optimized software installation and system configuration workflows, supporting lab sessions for 200+ students per semester through standardized maintenance procedures.

Research & Development Assistant — HIMASI — Universitas Nasional Jun 2024 – Mar 2025

- Orchestrated a national technology seminar as Chief Executive, coordinating 15 committee members from planning to execution and successfully delivering the event to 150+ attendees on schedule.
- Developed and managed the association's official WordPress website, increasing website traffic by 40% and driving member engagement through structured, SEO-optimized content.

Fullstack Developer — UNAS FEST — Universitas Nasional Festival Apr 2024 – Nov 2024

- Developed 10+ responsive website components using TypeScript and Tailwind CSS, achieving a Lighthouse score of 90+ for performance and accessibility on the official festival portal.
- Built Next.js API Routes for attendee registration and database integration, and engineered a testing pipeline with Jest and React Testing Library that slashed production bug rate by 60%.

EDUCATION

Universitas Nasional 2022 – 2026

Bachelor of Computer Science (S.Kom) — Information Systems | GPA: 3.77

TECHNICAL SKILLS

FRONTEND

React Next.js TypeScript Tailwind CSS

HTML / CSS

BACKEND & DB

Prisma Redis Node.js Express.js MongoDB

MySQL / SQL Postman

DATA & ANALYSIS

Python Pandas NumPy Matplotlib Tableau

Streamlit

MACHINE LEARNING

Scikit-learn XGBoost LSTM / Keras TensorFlow

DEVOPS & TOOLS

Jest & RTL GitHub Actions Git & GitHub

Google Cloud Docker Figma Jupyter Notebook

CERTIFICATIONS

Junior Web Developer

BNSP (Badan Nasional Sertifikasi Profesi) · Feb 2026

IT Specialist — Python

Certiport (Pearson VUE) · Oct 2025

Scrum Mastery

Telkom Indonesia · Jul 2025

Internet of Things Fundamental

Telkom Indonesia · Apr 2025

Statistics Using R for Data Science

DQLab · Feb 2023